

GANGADHAR BEHERA

AI & IoT Developer | Web & Software Developer | Computer Science Student

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PROFESSIONAL PROFILE

Enthusiastic and practical Computer Science undergraduate with focused hands-on expertise across Artificial Intelligence, Computer Vision, Embedded IoT architectures, and full-stack software development. Proven ability to build intelligent systems connecting hardware sensors (ESP32, LoRa) with computer vision (OpenCV, YOLO) and responsive software tools. Driven to engineer reliable, innovative solutions for complex real-world challenges.

TECHNICAL SKILLS

Programming:	Python, C, C++, Java, JavaScript, HTML5, CSS3
AI & Computer Vision:	OpenCV, YOLO, Object Detection, Machine Learning, Deep Learning Architectures
IoT & Embedded:	Arduino, ESP32, ESP8266, Raspberry Pi, LoRa, MQTT, Sensor Telemetry, Hardware Prototyping
Applied Systems & Tools:	Computer Applications (PGDCA), Financial Accounting & Computing (DCFA), Database Management (MySQL), System Operations, Business Data Analysis, Git, GitHub, VS Code, Arduino IDE
Currently Exploring:	React, Node.js, MongoDB, Firebase, Cloud Services (AWS, GCP), Figma

KEY ENGINEERING PROJECTS

NETRA-MINE — Intelligent Low-Visibility Mine Vehicle Navigation | Team *THE SMART POINTERS*

- **Overview:** Engineered an advanced situational awareness and safety telemetry prototype for open-cast mine vehicles operating in heavy fog, mist, and dust.
- **Hardware & Telemetry:** Deployed ESP32 microcontrollers with long-range LoRa wireless transceivers and obstacle sensors to stream real-time collision hazards without relying on public cellular infrastructure.
- **Vision System:** Incorporated computer vision processing modules for edge vehicle detection and low-light visual enhancement.

OPTISAFE-AI — Computer Vision Industrial PPE Compliance System | *Python, OpenCV, YOLO, Deep Learning*

- **Overview:** Developed an intelligent video safety pipeline designed to automatically verify personal protective equipment (PPE) compliance across hazardous worksites.
- **Vision Pipeline:** Utilized custom-trained YOLO object detection models to identify safety helmets, high-visibility vests, and unauthorized zone breaches in real-time camera streams.
- **Alerts & Logging:** Integrated automatic visual alerting, bounding box overlays, and incident reporting logs for safety officers.

EDUCATION

B.Tech — Computer Science & Engineering

Nalanda Institute of Technology, Bhubaneswar

Higher Secondary (+2 Science)

Talanagar Higher Secondary School

Secondary Education (10th)

Sunari High School, Sunari

Current • 2nd Year (3rd Sem)

Affiliated to BPUT, Odisha

2021 — 2023 • 77%

CHSE Odisha

2021 • 84%

BSE Odisha